PrepAIred: A Reinforcement-Learning-Adapted, Evidence-Gated Technical Interview Preparation System

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Abstract

Technical interview practice platforms require evaluating technical reasoning and verbal communication under realistic conditions. However, automated systems typically collapse delivery into technical scoring or rely on unconstrained Large Language Models (LLMs) vulnerable to prompt drift, hallucinated criteria, and verbosity bias. We present PrepAIred, an interview-preparation architecture that decouples scoring from dialogue generation. PrepAIred combines a multi-tier technical evaluator (semantic similarity, rubric concept coverage, fine-tuned CrossEncoder reasoning), an independent acoustic confidence model, a continuous-state Proximal Policy Optimization (PPO) difficulty controller, and deterministic evidence gating that restricts the LLM to grounded follow-ups. We evaluate technical scoring against an exploratory human-reference benchmark of $n = 64$ constructed answers across eight computer science questions, reporting exact deployed scoring formulas and code-verified implementations. The CrossEncoder reasoning signal correlates with human judgment (Spearman $\rho = 0.4832$, 95% CI [0.250, 0.676]), and the deployed composite achieves $\rho = 0.3812$. However, a length-confound analysis reveals the CrossEncoder is strongly associated with answer length ($\rho = 0.6594$, 95% CI [0.457, 0.831]); controlling for length, its partial correlation drops to 0.2434 (95% CI [−0.008, 0.507]). Because this confidence interval spans zero, an independent reasoning contribution beyond answer length is not statistically established on this benchmark. Furthermore, we make no claims of candidate learning gains, and take no position on whether PPO outperforms simpler guardrailed policies.

Index Terms — Adaptive Systems, Natural Language Processing, Reinforcement Learning, Large Language Models, Technical Interview Evaluation.

I. Introduction

Technical interviews in computer science, data structures, algorithms, and system design require candidates to simultaneously demonstrate domain knowledge, analytical problem-solving, structured logic, and fluent communication. Mock interview practice is a standard part of interview preparation, but scalable practice platforms face a fundamental trade-off. Human mock interviews provide high-quality, contextual feedback but are costly, non-scalable, and subject to evaluator variability. Automated preparation systems traditionally rely on static question repositories with pre-scripted test cases, failing to adapt dynamically when a candidate shows partial conceptual understanding, misunderstands a core requirement, or demonstrates strong technical correctness alongside hesitant verbal delivery.

The recent emergence of general-purpose Large Language Models (LLMs) has enabled conversational interview agents capable of generating dynamic questions and natural-language feedback. Despite their fluency, unconstrained LLM interviewers introduce operational risks in educational assessment: grading inconsistency, prompt drift, hallucinated rubric criteria, and a tendency to reward polished phrasing over technical correctness — unconstrained prompts can let a candidate achieve a favorable evaluation through keyword repetition even when the underlying reasoning is flawed. Standalone LLMs also lack principled difficulty adaptation, relying on subjective prompt interpretation rather than a structured, evidence-driven policy.

PrepAIred addresses these problems with a decoupled architecture that separates rigorous quantitative technical and speech assessment from qualitative natural-language generation, and strips the LLM of scoring authority entirely. This paper documents the system as it is actually implemented, and reports — favorably and unfavorably — what the project's evidence base does and does not support.

Contributions, scoped to what the evidence in this paper actually establishes:

1. A fully specified, code-verified system architecture for live, multi-turn, spoken technical-interview practice that structurally separates deterministic, multi-signal scoring from LLM-generated dialogue: the LLM is confined to phrasing grounded follow-ups and feedback under evidence-based triggers it does not control, and difficulty is adapted from a continuous six-dimensional state that treats speech-derived confidence as a first-class, non-penalizing signal rather than folding it into the technical score. A focused literature comparison (§II) finds this specific combination — applied to a live spoken interaction with a human candidate, with an RL-based continuous-state difficulty controller — was not present in the closest comparable systems identified; this is reported as an architectural and implementation-verification contribution, not a claim of demonstrated superiority over an LLM-judge baseline, which this paper does not test.
2. A precisely specified, code-verified multi-tier technical evaluator combining semantic similarity, rubric-based concept coverage, and a fine-tuned CrossEncoder reasoning signal, with every published weight and threshold matching the deployed implementation exactly (§IV).
3. A precisely specified, code-verified independent acoustic/temporal communication-confidence model, decoupled from technical scoring with one disclosed, narrow exception (§V).
4. A deterministic, evidence-gated follow-up and feedback mechanism, described here exactly as implemented (§VII).
5. A rigorous, honest characterization of the CrossEncoder scoring signal's actual evidentiary status, including a length-confound analysis showing its correlation with human judgment is substantially explained by answer length, with a length-independent contribution not established on the current benchmark (§X.B) — reported as a disclosed limitation, not implied validation.
6. A per-signal association analysis of the three evaluator signals' own stored scores against the same human-reference benchmark (not a component ablation — no signal was removed from the deployed composite and recomputed), reporting each signal's own correlation with human judgment separately: the CrossEncoder signal's is confidently non-zero, the semantic-similarity signal's is not, and the concept-coverage signal's is boundary-ambiguous across two independently frozen bootstrap analyses (§X.B, Table V) — evidence not previously reported for this system, computed from already-frozen data rather than a new experiment; and a follow-up paired, question-

aware bootstrap contrast between the signals (§X.B, Table VI) showing none of the four tested pairwise gaps, including the deployed composite versus the CrossEncoder alone, is statistically distinguished from zero at this sample size — also computed from the same already-frozen data, exploratory and not preregistered.

7. Software-verification evidence (automated test suites, including a newly fixed evaluator-reliability defect) establishing implementation correctness — explicitly not a claim of pedagogical effectiveness, candidate learning gain, or evaluator accuracy (§X.A).

We do **not** claim that PPO measurably improves difficulty tracking beyond a simple guardrailed constant policy: no evidence currently eligible for citation in this paper establishes that contribution either way (§VI).

II. Related Work

Automated short-answer scoring and technical assessment have evolved from early surface-level keyword matching to modern transformer-based semantic representations. Sentence-BERT demonstrated that Siamese transformer networks can produce semantically meaningful sentence embeddings for fast similarity comparisons [1]. However, semantic similarity alone remains vulnerable to false positives when candidate responses contain relevant terminology without correct logical structure. CrossEncoder models trained on passage-ranking datasets such as MS MARCO provide finer-grained pairwise entailment verification [2]. Prior automated assessment research has focused primarily on static answer grading; PrepAIred integrates a CrossEncoder-based reasoning score into a broader adaptive interview system that connects quantitative scoring directly to downstream question selection and grounded follow-up generation.

Speech analysis and communication modeling in professional interviews have established that non-verbal prosodic features correlate with perceived confidence and interview readiness. Classical acoustic toolkits such as Praat enable extraction of physical voice parameters including jitter, shimmer, and Harmonics-to-Noise Ratio (HNR) [3]. Subsequent interview-prediction systems combined language representations with acoustic parameters to predict human ratings [4], and automated interview-video frameworks applied multimodal regression to evaluate candidate delivery [5]. Existing speech-enabled interview tools frequently fold acoustic metrics into a single overall performance score, which can penalize technically proficient but vocally hesitant candidates. PrepAIred instead maintains speech confidence as an independent, non-penalizing state dimension that informs difficulty selection without degrading technical correctness scores.

Reinforcement Learning (RL) and Intelligent Tutoring Systems (ITS) have long addressed adaptive pedagogical sequencing. Early cognitive tutors modeled student skill acquisition using Bayesian Knowledge Tracing [6] and rule-based cognitive architectures [7]. Curriculum reinforcement learning formulates educational path optimization as a Markov Decision Process, using policy-gradient algorithms such as Proximal Policy Optimization (PPO) to balance challenge and capability [8], and automated-curriculum methods generalize this idea to broader RL settings [9], [10]. Adaptive testing frameworks grounded in item response theory similarly aim to optimize test length and difficulty [11]. Where traditional RL tutoring adjusts content across structured, static lessons, PrepAIred applies PPO difficulty adaptation within a multi-turn, speech-interactive technical interview environment governed by a continuous six-dimensional observation state — with the caveat, disclosed in §VI and §XII of this paper, that no evidence eligible for this paper establishes that this adaptation improves on a simpler guardrailed policy.

LLMs in educational assessment offer flexible narrative feedback but are documented to suffer from unconstrained generation and hallucination risk [12]. In technical-interview simulation specifically, unconstrained LLM interviewers can drift from rubric definitions or fail to enforce consistent difficulty escalation. PrepAIred addresses this by stripping the LLM of scoring authority entirely, using a locally served Qwen2.5 model (§III) exclusively as a grounded natural-language generator operating strictly under deterministic, evidence-driven state-machine triggers, with model outputs constrained by the Qwen technical report's own documented instruction-following behavior [13].

Multimodal interview-simulation frameworks have recently been extended toward broader career-guidance problems: Ascend AI provides a multimodal architecture unifying psychometric candidate profiling, predictive career-path modeling, personalized learning-roadmap generation, and interview simulation [14]. PrepAIred differs in scope: rather than end-to-end career guidance, it focuses specifically on single-session adaptive technical-interview assessment and difficulty optimization, combining question-specific hybrid answer evaluation, an independent acoustic communication-confidence model, PPO difficulty adaptation over a continuous six-dimensional state, and evidence-gated deterministic follow-up selection.

Closer to PrepAIred's own setting, Gomez et al. report a formative qualitative study (n = 20) of an AI-driven mock technical-interview tool with whiteboarding tasks and real-time feedback, finding participants described the experience as realistic and reported increased confidence, alongside challenges with conversational flow and timing [15]. That system does not report a reinforcement-learning difficulty controller or a deterministic scoring component decoupled from the dialogue model; confidence there is a self-reported qualitative outcome rather than a modeled, acoustic system state feeding a difficulty policy, which is the role it plays in PrepAIred (§V, §VI). A more direct validation of this paper's evidence-gating principle, in a different setting, comes from work compiling a rubric into a locked, executable specification and enforcing evidence-anchored, structured LLM-judge scoring for essay- and summarization-style text evaluation, improving alignment with human scores and stability under rubric-phrasing variation [16]. That result is independent evidence that evidence-anchored scoring can measurably improve over unconstrained LLM judgment in the domain it was tested in; PrepAIred applies a structurally similar separation of scoring authority from the dialogue model to a live, multi-turn, spoken interaction with a human candidate rather than static text evaluation, and reports its own evaluator-correlation evidence (§X.B) rather than importing that result. A third related direction lets one LLM play the interviewer role to dynamically evaluate another LLM across multi-turn question/feedback/follow-up exchanges, with no human candidate, no reinforcement learning, and no deterministic non-LLM scoring component — the interviewer LLM is simultaneously the questioner and the judge [17]. PrepAIred targets human candidates and removes scoring authority from the LLM entirely rather than assigning more of it. A focused search for a system combining all of PrepAIred's elements — a code-verified, deterministic multi-signal scorer decoupled from an LLM dialogue role; a live, spoken, multi-turn interaction with a human candidate; an independent acoustic confidence signal; and an RL-based continuous-state difficulty controller — did not find one; this is reported as the outcome of a focused comparison against the closest identified systems, not as a claim that no such system exists.

Table I synthesizes this comparison.

TABLE I — Comparative Analysis of Related Literature and PrepAIred

Research Theme	Representative Works	Typical Scope / Limitations	PrepAIred Approach
Automated Answer Assessment	Reimers & Gurevych [1], Bajaj et al. [2]	Static scoring, not embedded in an interactive questioning loop	Combines SBERT, rubric coverage, and CrossEncoder into an adaptive loop
Speech & Communication	Boersma [3], Naim et al. [4], Chen et al. [5]	Typically combine speech metrics into a single overall performance score	Separates speech confidence as an independent RL state dimension
Adaptive Tutoring & RL	Anderson et al. [7], Schulman et al. [8], Graves et al. [9]	Curriculum/tutoring RL commonly applied to static or structured tasks, not live multi-turn interviews	Applies PPO over a 6D state targeting technical interviews (tracking benefit not established here, §VI)
LLMs in Assessment	Ji et al. [12], Qwen Team [13]	Documented hallucination/inconsistency risk under unconstrained generation	Restricts the LLM to grounded follow-ups, gated deterministically
Multimodal Career Frameworks	Ascend AI [14]	Addresses end-to-end career guidance and learning-roadmap generation, not interview-answer evaluation	Focuses specifically on technical answer evaluation and 6D PPO adaptation
AI Mock Interview Tools	Gomez et al. [15]	n = 20 qualitative study; confidence is self-reported, not a modeled system state; no RL, no decoupled deterministic scorer	Models confidence as a computed acoustic/temporal signal feeding a difficulty controller
Evidence-Anchored LLM Judging	Hong et al. (RULERS) [16]	Validates rubric-locked, evidence-anchored scoring for static essay/summarization text evaluation; no interactive dialogue, no RL	Applies a structurally similar evidence-gating principle to a live, spoken, multi-turn human interview
LLM-as-Interviewer Evaluation	Kim et al. [17]	LLM interviews another LLM; no human candidate; the interviewer LLM is also the judge	Targets human candidates; removes scoring authority from the LLM entirely

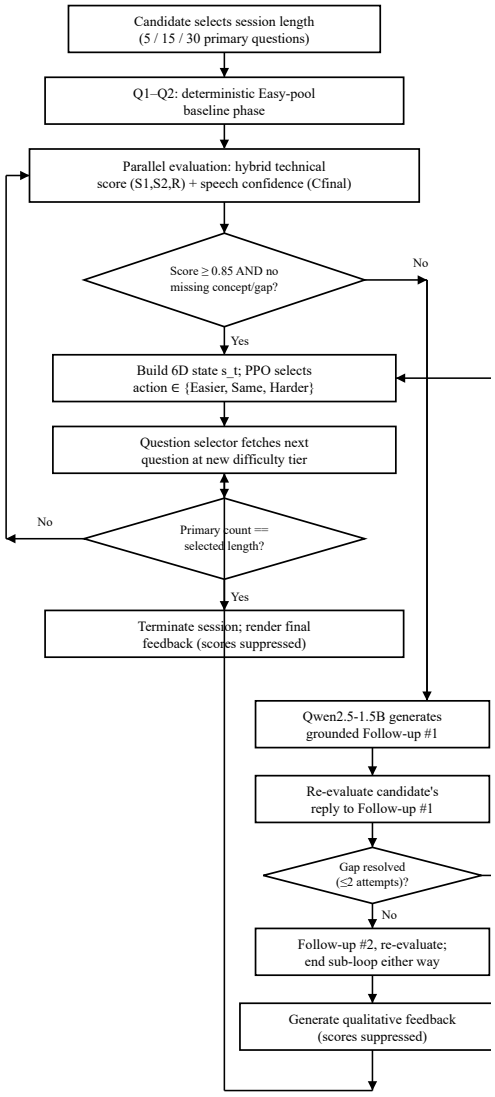
III. System Architecture

PrepAIred is built on a decoupled microservice architecture: a React 18 single-page application, a FastAPI/WebSocket session gateway, an asynchronous orchestration engine, a PyTorch technical evaluation service, an audio feature-processing module, a Stable-Baselines3 PPO difficulty controller, and a **local GGUF-quantized LLM runtime (llama.cpp, Qwen2.5-1.5B-Instruct, Q4_K_M)**. The LLM-serving module's startup routine (`services/qwen/app.py`) loads only Qwen2.5-1.5B-Instruct, first via GGUF/llama.cpp and, only if that fails and a research-GPU environment flag is explicitly set, via a PyTorch Transformers fallback of the same 1.5B model; it does not call any remote or Ollama-hosted service, and does not load a 7B model under any startup path in the current source — a "Qwen 7B" attribution label exists in the module's response schema for a research configuration, but no code path registers a 7B model, so that label is unreachable in the tested configuration. This section describes the deployed and tested configuration: Qwen2.5-1.5B-Instruct only.

The end-to-end session lifecycle (Fig. 1) is: session-length selection (5, 15, or 30 primary questions); a deterministic Easy-pool baseline phase for the first two questions; parallel technical/speech evaluation per turn; a deterministic follow-up decision layer (§VII); PPO-driven difficulty adaptation from the third question onward; and a final feedback view with all numeric scores suppressed from the candidate.

TABLE II — PrepAIred Architectural Component Specification

Component	Method / Library	Operational Role
Web Frontend	React 18 SPA + Web Speech API	User interaction, audio capture, score-suppressed UI
API Gateway	FastAPI + WebSockets	Real-time session routing and state orchestration
Hybrid Evaluator	SentenceTransformers + FAISS + CrossEncoder	Deterministic technical answer scoring
Confidence Engine	Praat-derived acoustic features + timing analysis	Speech delivery and vocal confidence modeling
RL Strategy	Stable-Baselines3 PPO	Continuous-state difficulty adaptation
LLM Engine	Qwen2.5-1.5B-Instruct via local GGUF (llama.cpp)	Grounded follow-up question and feedback generation
Question Bank	Structured JSON / FAISS indices	Rubrics, concept vectors, and question pools



Coarser two-condition-class gate, as implemented (§VII) — not the originally-designed five-condition trigger table.

Fig. 1. End-to-end operational workflow of PrepAIred. The candidate configures session parameters, establishing a two-question Easy-pool baseline phase. Text transcripts and speech audio undergo parallel evaluation to yield technical correctness (S_1, S_2, R) and communication confidence (C_{final}) scores. Deterministic evidence logic (§VII) determines whether to initiate a grounded follow-up probe (up to two bounded attempts)

or proceed directly through the 6D PPO state-adaptation loop (§VI) until the selected primary-question count is reached. *(This diagram describes control flow, cross-checked against the orchestrator code; it is not a data figure and is not regenerated from a results file.)*

IV. Hybrid Technical Answer Evaluation

The technical evaluation engine computes a score $S_{final} \in [0.0, 1.0]$ by comparing candidate transcript T_c against a canonical question rubric R_q, containing semantic reference statements $R_q^{semantic}$, required concepts $R_q^{concept}$, misconception vectors M_q, advanced-insight vectors B_q, and mandatory-concept flags.

A. Semantic similarity (S_1). T_c is segmented into sentences $\{s_1, \ldots, s_n\}$; each sentence and each reference statement $t_j \in R_q^{semantic}$ are embedded with `all-MiniLM-L6-v2` (384-dimensional). S_1 is the mean, over reference statements, of the maximum cosine similarity to any candidate sentence:

$$S_1 = \frac{1}{|R_q^{semantic}|} \sum_{j=1}^{|R_q^{semantic}|} \max_{1 \le i \le n} \cos(e_i, v_j). \tag{1}$$

B. Concept coverage (S_2) and dampening. For each required concept $c_k \in R_q^{concept}$, presence is an indicator at cosine-similarity threshold $\tau_{concept} = 0.30$:

$$S_2 = \frac{1}{|R_q^{concept}|} \sum_{k=1}^{|R_q^{concept}|} \mathbb{1}\Big(\max_{1 \le i \le n} \cos(e_i, c_k) > 0.30\Big). \tag{2}$$

To resist keyword stuffing (terms present without connecting logic), effective coverage is dampened when reasoning score R is at or below a floor of 0.30:

$$S_2^{effective} = \begin{cases} S_2, & R > 0.30 \\ 0.60 \cdot S_2, & R \le 0.30. \end{cases} \tag{3}$$

C. CrossEncoder reasoning score (R). T_c and the reference answer A_q are jointly scored by `cross-encoder/ms-marco-MiniLM-L-6-v2`; the raw logit r_{raw} is linearly rescaled and clamped:

$$R = \text{clamp}\Big(\frac{r_{raw} - 0.20}{0.70},\, 0.0,\, 1.0\Big). \tag{4}$$

D. Bonus, mistake penalty, and mandatory concepts. Mistake penalty accumulates matched misconception-vector hits (threshold 0.55, increment 0.07, capped at 0.30):

$$P_{mistake} = \min\Big(0.30,\; \sum_{m \in M_q} 0.07 \cdot \mathbb{1}(\max_i \cos(e_i, m) > 0.55) + P_{reasoning}\Big), \tag{5}$$

bonus credit accumulates matched advanced-insight hits (threshold 0.50, increment 0.03):

$$S_{bonus} = \sum_{b \in B_q} 0.03 \cdot \mathbb{1}(\max_i \cos(e_i, b) > 0.50). \tag{6}$$

The base score aggregates the three weighted signals, then bonus/penalty are applied and clamped:

$$S_{base} = 0.15\, S_1 + 0.35\, S_2^{effective} + 0.50\, R, \qquad S_{final} = \text{clamp}(S_{base} + S_{bonus} - P_{mistake},\, 0.0,\, 1.0). \tag{7}$$

If mandatory concepts are required and any is missing, the score is hard-capped: $\text{Mandatory Pass} = \text{False} \implies S_{final} = \min(0.60, S_{final})$.

TABLE III — Hybrid Technical Evaluator Parameter Configurations (every value below was re-verified against `services/evaluator/app.py` and matches exactly)

Parameter	Value	Description
Semantic weight (S_1)	0.15	Weight on SBERT similarity
Concept weight (S_2)	0.35	Weight on structural concept coverage
Reasoning weight (R)	0.50	Weight on CrossEncoder reasoning
Concept threshold ($\tau_{concept}$)	0.30	Cosine cutoff for concept match
Reasoning floor	0.30	CrossEncoder threshold triggering dampening
Dampening factor	0.60	Concept-coverage multiplier when $R \le 0.30$
Bonus increment / threshold	0.03 / 0.50	Credit per matched insight / match cutoff
Mistake increment / threshold	0.07 / 0.55	Penalty per matched mistake / match cutoff
Penalty cap	0.30	Maximum accumulated mistake penalty
Mandatory cap	0.60	Maximum score if mandatory concepts missed

Every formula and constant above was independently verified against the deployed code (`services/evaluator/app.py`) and matches exactly.

V. Communication Confidence Model

Candidate vocal delivery is modeled independently of technical correctness, so that hesitation or delivery nervousness does not alter technical scores while still letting the RL policy adapt to candidate confidence state.

A. Prosodic voice quality (V_Q). From pitch jitter j, shimmer s, Harmonics-to-Noise Ratio h, and pitch-variance C_{pitch_var}:

$$C_{jitter} = \text{clamp}(1 - j/0.03, 0, 1), \quad C_{shimmer} = \text{clamp}(1 - s/0.12, 0,$$

$$V_Q = 0.32\, C_{jitter} + 0.30\, C_{shimmer} + 0.28\, C_{HNR} + 0.10\, C_{pitch_var}. \tag{9}$$

B. Fluency, rate, rhythm. Fluency penalizes excess pause time (t_{pause}/t_{speech}, scaling 1.7); rate fluency rewards words-per-minute in $[2.1, 4.3]$ (center 3.2) and decays outside that band; rhythm penalizes more than 7 detected pauses (scale 12) — formulas verified against `agents/audio/confidence_scorer.py`.

C. Composite confidence (C_{final}).

$$C_{final} = 0.28\, V_Q + 0.27\, L + 0.22\, F_{fluency} + 0.13\, F_{rate} + 0.10\, F_{rhythm}$$

with L a linguistic-confidence signal from ASR output; if ASR confidence $A_{conf} < 0.45$, the model shifts weight toward the acoustic signal: $C_{final} = 0.75\, C_{final} + 0.25\, V_Q$. Numerical confidence values are never shown to the candidate.

A narrow qualification to independence. Vocal hesitation does not alter technical scores through the acoustic channel — C_{final} has no parameter in the technical evaluator. Independence is not unconditional, however: the technical evaluator applies a small, capped penalty (0.03) when the candidate's **transcript text** contains a lexical hedging word (e.g. "maybe," "not sure," "umm") **and** the answer's content is independently weak (concept coverage < 0.50 and reasoning < 0.40). This is a text-level signal, not the acoustic model, and only ever applies alongside already-weak content.

VI. Reinforcement Learning for Adaptive Difficulty

PrepAIred formulates difficulty selection as a Markov Decision Process solved with PPO. The state space, action space, and transition rules below were independently confirmed against the project's canonical specification.

TABLE IV — CONTINUOUS SIX-DIMENSIONAL RL STATE SPACE

Index	Variable	Range	Definition
0	Performance (P_t)	[0,1]	Current-turn technical score S_{final}
1	AvgPerformance ($\bar{P}_t$)	[0,1]	Rolling mean of technical scores
2	Confidence (C_t)	[0,1]	Composite speech confidence C_{final}
3	Hesitation (H_t)	[0,1]	$1 - F_{fluency}$
4	TimeNorm (T_t)	[0,1]	Elapsed session time / budget
5	Difficulty (D_t)	[0,1]	Current difficulty tier / 5

The agent selects a discrete action $a_t \in \{0, 1, 2\}$ (Easier / Same / Harder), moving difficulty tier $D_t \in \{1, \ldots, 5\}$ by ∓ 1 or 0, bounded to that range.

Reward formulation. The frozen PPO checkpoints evaluated in this paper were trained against the project's canonical training configuration: a weighted reward combining decision alignment with a hand-coded rule oracle (weight 0.60), an outcome-delta term (0.30), a multimodal shaping term (0.10), and a stability penalty (−0.10). The project's own training records describe this reward as "largely imitation of a rule oracle" rather than target-tracking against a fixed performance setpoint.

On PPO's own effectiveness, this paper takes no position, because no evidence eligible for citation here establishes one. The frozen PPO checkpoints were trained against a single default simulated candidate persona, with no persona sampler in the training scripts, and the training candidate's noise generator is unseeded, so no claim that these checkpoints generalize across candidate personas, or that the training seeds alone would reproduce them, is supported. Beyond that, this paper does not assert that PPO improves difficulty tracking over simpler alternatives, because it has no evidence of its own, and no evidence yet eligible for citation, that would support such a claim in either direction. A separate, not-yet-public companion manuscript is the dedicated subject of PPO effectiveness for this system; under this project's citation-timing policy, its findings are not used here, even as uncited narrative summary.

VII. Deterministic Follow-Up and Feedback

PrepAIred decouples the decision to initiate a follow-up probe from natural-language generation: follow-up probing is governed by deterministic evaluator evidence, not unconstrained LLM judgment.

Trigger logic (derived directly from `agents/orchestrator/interview_orchestrator.py` and `services/evaluator/app.py`):

- A follow-up is **skipped** only when the technical score is ≥ 0.85 **and** there is no missing

concept, no detected misconception, and no other identified gap.

- Otherwise, a follow-up is **triggered** whenever there is a missing concept, a detected

misconception, a technical score below 0.80, or any other identified gap — including a failed mandatory-concept check or reasoning below 0.50, evaluated in that priority order.

- At most two follow-up attempts are permitted per question.

This mechanism is coarser than the project's originally-designed five-condition trigger table (which included a separate high-score "Depth Probe" intended to surface edge cases); the deployed code implements only the two-condition-class rule above, and does the **opposite** of a high-score probe — a high, clean score exits the follow-up loop rather than triggering one (see §XII). The deterministic, non-LLM-judged gating principle itself — evidence decides whether to probe, not LLM discretion — is what this paper reports as implemented and verified; the more elaborate five-condition design is not, since the code does not implement it.

When a trigger fires, the locally served Qwen2.5-1.5B-Instruct model generates a targeted follow-up prompt, grounded with the original question, the candidate's response, missing concepts, and identified misconceptions. If a gap persists after Follow-up #1, Follow-up #2 is generated with accumulated history; if unresolved after two attempts, the sub-loop terminates, qualitative feedback is generated, and the primary interview workflow resumes. The candidate's own transcript is displayed for reference, followed by six qualitative feedback sections (What You Got Right; Concepts, combining covered and missing concepts in one view; Incorrect or Incomplete; How to Improve; Delivery & Communication; AI Summary), plus a conditional Retry/Best-Answer Status section shown only when a prior comparable attempt exists; all quantitative scores (S_{final}, S_1, S_2, R, C_{final}) are strictly suppressed from this view, including an explicit stripping pass that removes any numeric score pattern the narrative-generation step might otherwise leak. This section structure and the score-suppression behavior were confirmed directly against the frontend feedback-rendering component (`apps/web/src/FeedbackCard.jsx`), correcting an earlier, less precise description of the feedback view (which had listed a different section breakdown not matching this component).

VIII. Question Selection and Session Control

Candidates select a primary session length of 5, 15, or 30 questions. The question-selection engine manages retrieval across Easy ($D \in \{1, 2\}$), Medium ($D = 3$), and Hard ($D \in \{4, 5\}$) pools. Questions 1 and 2 form a deterministic baseline phase, sampled independently from the Easy pool subject to an intra-session non-repetition constraint; PPO-driven adaptive selection activates from Question 3 onward. Follow-up questions are subordinate probes and do not increment the primary question counter (`primary_question_index` in `agents/orchestrator/interview_orchestrator.py` counts primary questions only, explicitly carrying the parent primary question's index for any follow-up); the session terminates when the primary count reaches the candidate-selected session length (`num_questions`, defaulting to 15 if unset, governs the primary-question budget the session runs against).

IX. Hyperparameter Selection

The evaluator's thresholds and weights (Table III) reflect the authors' design choices. No tuning log, development-set result, or sweep record documenting a systematic empirical tuning process was found in the project's records, so this paper does not claim that a recorded tuning study established these specific values, even though the constants themselves are verified correct against the deployed code (§IV).

X. Experimental Methodology and Verification

A. Implementation verification and source snapshot

This paper describes a specific version-controlled development snapshot, including the evaluator-reliability fix described below. The fix was covered by 16 targeted regression tests across its dedicated test files (`tests/unit/test_score_validator.py` and `tests/unit/test_evaluator_infrastructure_failure_flag.py`); all 16 passed on the documented working tree. The cases cover non-finite evaluator scores, missing (`None`) scores, non-numeric values, invalid objects, a contradictory upstream success status paired with invalid scores, and preservation of a genuine finite zero score. This is implementation-correctness evidence only: it does not establish evaluator validity, candidate learning, or improved interview outcomes. The fix was present in the reviewed working tree and had not yet been merged into the project's main integration branch at the recorded snapshot; exact revision details are retained in the project engineering record.

B. CrossEncoder reasoning model

Throughout this subsection, the **human-reference score** for each of the 64 constructed answers is the mean of three recorded ratings for 54 of the 64 answers, and an adjudicated score for the remaining 10 (the 8 keyword-stuffed cases and 2 partial-answer cases, where the three ratings diverged); it is not a uniform three-rater consensus across all 64 items, and it is described here as an exploratory benchmark, not an expert panel or independent adjudication — rater identity, qualifications, independence/blinding beyond the rating sheet content, consent, and any institutional ethics determination are not documented in the project's records (§XIII). The frozen source file's own internal label for the adjudicated subset is a file identifier, not a methodological claim, and is not reused as a description here. The three raters' inter-rater reliability on this benchmark is high (ICC(2,1) = 0.953, independently re-derived via a second, differently structured implementation, an exact match to the frozen figure) but this figure should not be read as evidence of fine-grained rater discrimination: a one-way ANOVA of each rater's raw score against the benchmark's 10 author-assigned construction categories finds 98–99% of the variance (η^2 = 0.9815, 0.9881, and 0.9849 for the three raters individually; 0.9891 for the gold score) in every rater's score, and in the gold score itself, is explained by which category the item belongs to, with the average disagreement between raters on the same item (SD ≈ 0.068) roughly 5–6× smaller than the spread between different items' category-driven means (SD ≈ 0.361). The high reliability substantially reflects a benchmark built from clearly separated quality categories (a correct concise answer is easy for any rater to distinguish from a keyword-stuffed one), not necessarily fine-grained agreement on more ambiguous within-category quality differences.

The frozen, comparable result is the R-only (CrossEncoder-only) Spearman correlation against the project's human-reference benchmark: **ρ = 0.4832**, case-bootstrap 95% CI [0.250, 0.676], n = 64 constructed answers across 8 questions — exploratory/initial evidence, not a general validity claim. The deployed evaluator is a **composite** of three signals (§IV), whose combined score has a separately measured, more thoroughly characterized correlation with the human-reference benchmark: Spearman ρ = 0.3812, case-bootstrap 95% CI [0.158, 0.577], same 64 items — numerically lower than the CrossEncoder-alone figure. A paired bootstrap contrast of this specific gap, reported below, finds it is not statistically distinguished from sampling noise at this sample size, so we describe the composite as numerically, not confirmedly, lower than R alone; the composite's design — discounting S_2 when R is low — remains a structural robustness rationale rather than an independently validated correlation trade-off (§XI).

Per-signal association analysis. The project's frozen scoring dataset for this benchmark stores each raw evaluator signal's own score alongside the deployed composite, letting this paper report each signal's own correlation with human judgment on the identical 64 items. This is **not a component ablation**: no signal was removed from the deployed composite and the composite recomputed without it; these are simply the already-stored, already-scored S_1, $S_2^{effective}$, and R values, each independently correlated against the human-reference gold. It is also not a new experiment: the underlying scores and their two-level cluster-bootstrap confidence intervals were already frozen under the project's internal preregistration gates, dated 2026-09-19. Every value below was independently reproduced twice — once by recomputing the point-estimate Spearman correlations directly from the stored scores, and once by re-executing the exact documented two-level bootstrap (`numpy.random.default_rng(42)`, B = 10,000, resample 8 questions with replacement then resample each drawn question's 8 answers with replacement, so the resampling unit is the question, respecting the 8-question clustering) — both reproductions match the frozen result exactly.

TABLE V — PER-SIGNAL ASSOCIATION WITH HUMAN JUDGMENT (TWO-LEVEL CLUSTER BOOTSTRAP, B = 10,000, SAME 64 ITEMS / 8 QUESTIONS AS §X.B)

Signal	Spearman ρ	95% CI
S_1 alone (SBERT semantic similarity)	0.2070	[−0.0231, 0.5027]
$S_2^{effective}$ alone (concept coverage, post-dampening)	0.3006	[−0.0057, 0.6523]
R alone (CrossEncoder)	0.4832	[0.2453, 0.6956]
Deployed composite ($0.15 \cdot S_1 + 0.35 \cdot S_2^{effective} + 0.50 \cdot R$)	0.3812	[0.1529, 0.6473]

The composite row is the same 0.3812 point estimate reported earlier in this subsection, not a new or separate measurement; its confidence interval here differs slightly from the [0.153, 0.649] two-level- cluster interval reported below (§X.B, "Composite-score robustness") because that figure comes from a second, independently-implemented frozen bootstrap (same nominal method and seed, a different underlying scoring pipeline) — both are legitimate, frozen results for the identical point estimate, and neither is more "correct" than the other.

This second frozen bootstrap also matters for $S_2^{effective}$ specifically: the second pipeline's independently computed two-level CI for $S_2^{effective}$ is [0.0031, 0.6543] — excluding zero — while the first, shown above, is [−0.0057, 0.6523] — including zero. The two point estimates agree (0.3006 in both), but whether S_2's CI excludes zero depends on which of two equally-frozen, equally-legitimate bootstrap runs is consulted; the true interval is evidently very close to this boundary. We therefore do **not** claim $S_2^{effective}$'s marginal correlation is confidently non-zero, but we also do not claim it is confidently zero-or-indistinguishable-from-zero — the data at this sample size cannot settle it either way, and it would be a similar overreach to pick whichever of the two frozen analyses gives the cleaner-sounding answer.

S_1's CI excludes a confidently non-zero reading under both frozen analyses ([−0.023, 0.503] and [−0.023, 0.501]), and R's CI excludes zero under both ([0.245, 0.696] and [0.246, 0.698]). **This is a set of four separate per-signal statements, each compared only against zero — it is not by itself a between-signal significance test.**

Paired between-signal contrast. An exploratory secondary analysis computes the actual paired, question-aware contrast that a marginal, per-signal comparison alone cannot provide. It reuses the same already-frozen, already-seen scoring data (no new model run, no new data) and the same two-level cluster bootstrap method and seed as the original per-signal analysis, but resamples both signals in a pair from the *same* bootstrap draw in every replicate, so the comparison is paired rather than two independent CIs. Method fixed in advance in a dated protocol note written before computing any result. All four confidence intervals were independently re-derived a second time via a differently-structured implementation (separate data-loading and statistics libraries, a separately-written resampling loop) against the same source data; all four reproduce exactly, confirming the pairing (the same resample applied to both signals each replicate) is implemented correctly and not merely internally consistent. This remains an **exploratory, inconclusive paired comparison of already-seen, already-frozen data** — not a preregistered confirmatory test, and its results below are not read as evidence that the signals are statistically equivalent.

TABLE VI — Paired Signal Contrasts (two-level cluster bootstrap, B = 10,000, seed 42, same draw for both signals per replicate; exploratory, not preregistered)

Contrast	Point difference (ρ)	95% CI	Excludes zero?
$R - S_1$	+0.2761	[−0.0747, 0.5745]	No
$R - S_2^{effective}$	+0.1826	[−0.1632, 0.5353]	No
$S_1 - S_2^{effective}$	−0.0936	[−0.2938, 0.1654]	No
Composite − R alone	−0.1020	[−0.2853, 0.1171]	No

None of the four paired contrasts excludes zero. Despite R's marginal correlation (0.4832) being more than double S_1's (0.2070), the paired bootstrap does not distinguish R from S_1 or from $S_2^{effective}$ at this sample size (n = 64 cases, 8 question clusters); nor does it distinguish S_1 from $S_2^{effective}$; nor does it distinguish the deployed composite from R alone, despite the composite's numerically lower point estimate. This does not mean the signals are equivalent — a CI that includes zero at n = 64/8 is underpowered evidence either way, not evidence of no difference — but it does mean the earlier text's caution was correct, and is now a tested rather than merely a withheld claim. We therefore continue to explicitly not claim R is uniquely or significantly better than S_1 or S_2, and now additionally do not claim the deployed composite is a statistically confirmed correlation trade-off relative to R alone — only that its point estimate is numerically lower. We do not conclude S_1 or S_2 are non-functional in the deployed system — they serve structural roles (keyword-stuffing resistance via dampening, mandatory-concept gating) that a bare correlation with an overall human quality rating does not capture. This is a current-cycle, non-P2 result and must not be confused with the project's separate, paused, N=20-pilot seven-way evaluator ablation, which used a different, superseded, single-rater human-reference set, genuinely recomputed the composite under several signal-combination weightings (a real ablation grid), and stays excluded from this paper (§XII).

R's own marginal association (both frozen analyses agree it excludes zero) must be read together with, not instead of, the length-confound analysis immediately below: that analysis finds R's correlation with human judgment is substantially explained by its own association with answer length, and the length-controlled partial correlation's 95% CI includes zero. We do not describe R as validated "reasoning verification" on the strength of this section's marginal correlation — a non-zero marginal association is not the same claim as an established, length-independent reasoning contribution, and the latter is the one this paper does not make.

Composite-score robustness to resampling (about the full composite score, not the CrossEncoder component alone). A two-level (question-then-case) cluster bootstrap, B = 10,000, widens the interval to [0.153, 0.649], reflecting the coarser 8-question effective sample; leave-one-question-out correlations stay in a narrow band (0.352–0.434); leave-one-rater-out correlations are similarly stable (0.355–0.382).

CrossEncoder component vs. simple lexical baselines. A preregistered exploratory analysis arm (tag `prereg/X2-B/v2`) compared the CrossEncoder's own correlation against a trivial word-count-only baseline, using the **raw, unmapped CrossEncoder logit** (ρ = 0.4825) rather than the deployed, clamped-and-rescaled R signal reported above (ρ = 0.4832): **ρ = 0.4897, 95% CI [0.219, 0.708]** — matching or exceeding the CrossEncoder's own correlation. BM25 (ρ = 0.3812), TF-IDF (ρ = 0.245), and reference-token overlap (ρ = 0.4301) are in a similar range. The 0.4825/0.4832 difference is not resampling noise — both are fixed point estimates on the same 64 items — and is fully explained: applying the deployed mapping formula to the raw logit reproduces the stored R value to within 0.0005 per item, and the residual is entirely storage-precision rounding (to 3 decimals) that creates two tied pairs among 64 items, shifting 4 items' ranks by a fraction of a rank and the resulting Spearman ρ by 0.0007; independently verified by tracing the deployed mapping code against the stored values. This was that arm's registered *exploratory*, not primary confirmatory, endpoint — a planned, pre-run-noted analysis, not post-hoc fishing, but not designed or powered as a formal head-to-head test, so its result is suggestive, not decisive. Read plainly: the CrossEncoder's advantage over a naive length/lexical baseline is not established on this benchmark.

Length-confound analysis. This motivated a more direct question: does the CrossEncoder carry information about human judgment beyond answer length itself? A partial-correlation analysis (method fixed in a pre-run note before computation; exploratory/sensitivity status; same 64-case benchmark) is computed on the same **raw, unmapped CrossEncoder logit** used immediately above (not the deployed R signal; the two differ only by the monotonic clamp/rescale mapping and storage rounding detailed above, which does not materially affect a rank-based partial-correlation analysis) and found the CrossEncoder's own score is associated with length (Spearman ρ = 0.6594 against raw token count, 95% CI [0.457, 0.831] — this specific association, unlike what follows, is confidently established on this benchmark; it is an association, not a demonstrated causal mechanism). Controlling for length, the CrossEncoder's partial correlation with human judgment drops from 0.4825 to **0.2434** — this drop is itself statistically supported (95% CI of the drop: [0.048, 0.429], excludes zero) — but the partial correlation's own remaining level has a 95% CI of **[−0.008, 0.507]**: the lower bound touches zero, so a length-independent contribution is not confidently established. This is not the same as establishing there is none — the upper bound (0.51) is equally consistent with a real, moderate contribution this 8-question sample cannot resolve. A direct, paired, question-aware comparison against BM25's own partial correlation (0.2458) gives a point-estimate difference of −0.0024 and a 95% CI of **[−0.341, 0.360]**: far too wide to conclude either that the CrossEncoder and BM25 differ or that they are equivalent once length is controlled for. We describe this component in this paper as a *reasoning-scoring signal whose independent contribution beyond length and simple lexical overlap is not established on the current benchmark*, not as validated "reasoning verification."

Separately, the same analysis arm found the project's own fine-tuned CrossEncoder correlates with the human-reference score better than the public, un-fine-tuned upstream checkpoint on identical inputs (paired Spearman difference 0.337, 95% CI [0.042, 0.606]) — real evidence the fine-tuning changed something, though exactly what cannot be fully characterized without the missing training record (§XII), and this does not by itself establish that what it learned is length-independent reasoning quality rather than, e.g., a stronger length-sensitivity than the upstream model.

We report the favorable robustness evidence and the unfavorable length-confound finding together, rather than either overstating the CrossEncoder's validation or omitting the null result. A benchmark specifically designed to separate reasoning quality from answer length is the concrete next step; it does not need to be designed from scratch — the project's existing, not-yet-registered X2-C confirmatory protocol already specifies six strata with length deliberately crossed with correctness, and a length-confound-specific analysis amendment to it is drafted. This is real, reviewable future work, gated on an institutional ethics determination not yet sought, and is not available before this paper's submission; the existing 64-case benchmark was not built for this purpose, and no further reanalysis of it will resolve the question with more precision than an 8-question effective sample allows.

XI. Results and Discussion

The hybrid evaluator's dampening mechanism (§IV) functions as specified and is code-verified. The communication-confidence model (§V) is fully code-verified and decoupled from technical scoring, with one disclosed, narrow lexical exception. The follow-up mechanism is deterministic and evidence-gated as intended, though its actual trigger logic is coarser than the system's originally-designed five-condition trigger table (§VII, §XII). Automated tests confirm implementation correctness across the evaluator, orchestrator, RL environment bounds, and audio pipeline (§X.A). We do not claim the PPO difficulty controller demonstrably improves question-

difficulty tracking over simpler alternatives: this paper has no evidence of its own on that question, and no evidence yet eligible for citation from elsewhere in the project establishes it either way. The composite evaluator's correlation with human judgment is moderately stable under resampling of questions and raters (§X.B). A per-signal association analysis (§X.B, Table V) finds R's own correlation with human judgment is confidently non-zero under two independently frozen bootstrap analyses, S_1's is not, and S_2's is ambiguous between the two — three separate per-signal readings. A follow-up paired, question-aware bootstrap contrast (§X.B, Table VI) tests the comparison directly and finds none of the four pairwise gaps — including R against S_1 or S_2, or the deployed composite against R alone — statistically distinguished from zero at this sample size; this is a disclosure worth making explicit about the evidentiary basis for the three components' 0.15/0.35/0.50 weighting rather than leaving it implicit, and it means that weighting is not itself justified by a demonstrated between-signal performance gap. The CrossEncoder (R) component specifically shows a real marginal correlation with human judgment, but the length-confound analysis found that correlation is substantially explained by the CrossEncoder's own strong association with answer length; a length-independent contribution is not confidently established (the CI's lower bound is near zero, not the point estimate), and a direct paired comparison against a simple lexical baseline (BM25) is too imprecise at this sample size to say whether the two differ. We report this as a genuine, currently unresolved limitation of the evaluator's reasoning-verification claim: the component functions as a reasoning-scoring signal whose independent value is not yet demonstrated, a materially different — and more honest — claim than "reasoning verification."

XII. Limitations and Threats to Validity

Local LLM inference introduces noticeable response latency during follow-up and feedback generation; speech-feature extraction is sensitive to background noise and microphone quality; technical evaluation precision depends on the completeness of curated question rubrics; and no human user study of learning outcomes has been run. In addition:

- The CrossEncoder reasoning model's training data, training code, and exact train/test split are not

established in the project's own records; its correlation with human judgment should be read as exploratory (n = 64 constructed answers, 8 questions), not general-purpose validity. The length-confound analysis (§X.B) found this correlation is substantially explained by the CrossEncoder's own association with answer length; a length-independent contribution is not confidently established — not the same as establishing there is none.

- This paper does not claim the PPO difficulty controller's own contribution to tracking accuracy

beyond a simple guardrailed constant policy, because no evidence eligible for citation here establishes it (§VI). The project's internal PPO research on this exact question belongs to a separate, not-yet-public companion manuscript and is withheld accordingly. The frozen PPO checkpoints were trained against a single simulated candidate persona with an unseeded noise generator, so no claim of cross-candidate generalization, or of exact checkpoint reproducibility from the training seeds alone, is supported.

- The 64-case benchmark's rater provenance and ethics records are incomplete; this benchmark must not

be described as independent, expert, or ethics-cleared human evaluation, only as an exploratory construction.

- The per-signal association analysis (§X.B, Table V) has the same 8-question effective sample size

as every other analysis on this benchmark; S_1's CI spanning zero, and $S_2^{effective}$'s boundary-ambiguous CI (excluding zero under one independently frozen bootstrap, including it under another), could reflect a genuinely weak marginal correlation or insufficient power to detect a real but modest one — the data available cannot distinguish these. A follow-up paired, question-aware bootstrap contrast (§X.B, Table VI) now exists between S_1, S_2, R, and the deployed composite; none of its four tested pairwise gaps excludes zero, so no claim that R differs significantly from S_1 or S_2, or that the composite differs significantly from R alone, is made — this is now a tested null result rather than an untested comparison, but a null result at n = 64/8 reflects limited power as much as it reflects a genuine absence of difference. The comparison is descriptive, with no multiplicity correction applied across the four rows of Table V or the four contrasts of Table VI.

- A defect in the follow-up/feedback generation path — a bare (non-list) evaluator field could

corrupt follow-up text into single characters — has been identified, fixed, and covered by regression tests on the development snapshot described in §X.A, but has **not yet been merged** into the codebase the deployed system currently runs from. Authors should confirm merge status before any submission that implies the fix is live.

- The deployed model checkpoint and the checkpoint referenced by this project's simulation evidence

are not confirmed to be the same file (different content hashes); no claim in this paper assumes they are identical, and no claim in this paper depends on which specific checkpoint is described.

XIII. Ethics, Provenance, and Data Statement

No new human-subjects data was collected for this paper. The 64-case benchmark used throughout §X.B was constructed and rated internally by the project team, following the project's own internal working practices. This paper's evidentiary characterization of that benchmark — exploratory, not an independent, expert-panel, or formally ethics-approved evaluation — describes the strength and scope of the resulting correlation evidence, not an assertion that the project's internal procedures were improperly followed; the project's own records do not include a separate, independently documented consent, blinding, or institutional-ethics record for this internal exploratory benchmark beyond its own working files, and this paper does not claim one exists. A confirmatory human-rated benchmark (X2-C) is designed but, as of the time of writing, not yet started under a separate, ongoing effort: no item has been authored for it, no rater has been contacted, and no data has been collected, pending an institutional ethics/venue determination that has been drafted but not yet sent. All simulated-candidate and PPO training evidence referenced in this paper (§VI, §XII) derives from a single synthetic simulated persona and synthetic reward signal; none of it is treated, described, or cited as evidence about real users, real candidate learning, or real interview outcomes.

XIV. Reproducibility and Data/Code Availability

PrepAIred is implemented in **Python 3.12** and React 18. The backend uses FastAPI, **PyTorch (pinned `>=2.3.0,<2.7.0`)**, SentenceTransformers, FAISS-CPU, Stable-Baselines3, and a **local GGUF/llama.cpp runtime** for LLM inference. All evaluator weights, thresholds, and model identifiers reported in this paper (§IV, Table III) are exact values from the deployed configuration, not illustrative examples.

Methodological reproducibility. Every statistical result reported in this paper was computed with a fixed, documented method and seed: the two-level (question-then-case) cluster bootstrap (`numpy.random.default_rng(42)`, B = 10,000) used throughout §X.B respects the benchmark's 8-question clustering by resampling questions first and then resampling each drawn question's answers; the length-confound analysis (§X.B) additionally used a rank-transform-and-residual-regression method, fixed in a written protocol note before computation; the paired signal contrast (§X.B, Table VI) used the same

two-level bootstrap with both signals resampled from the same draw per replicate, also fixed in a written protocol note before computation. Every load-bearing statistic in Tables V and VI, and in the length-confound analysis, was independently re-derived a second time via a differently structured implementation (a separate statistics library and a separately written resampling loop, or a mathematically distinct method), reproducing the original figures exactly or to within ordinary Monte Carlo noise — reported in this paper as independent verification of an existing calculation on the same already-collected data, not as independent replication on new data.

Data and code availability. The evaluator source code, RL training/environment code, and the 64-item benchmark dataset (human-reference scores, category labels, and per-signal scores) used throughout this paper are maintained in the authors' project repository. A public release of the repository, or of a de-identified subset sufficient to reproduce Tables III–VI, has not been finalized at the time of writing.

XV. Conclusion and Future Work

This paper presented PrepAIred's architecture as implemented and verified: a hybrid, precisely specified technical evaluator; an independent, precisely specified speech-confidence model; a deterministic evidence-gated follow-up mechanism, simpler than the system's originally-designed five-condition trigger table (§VII, §XII); and a PPO-based difficulty controller whose own marginal tracking benefit is not asserted here because no evidence eligible for citation in this paper establishes it. We report a length-confound analysis showing the CrossEncoder's correlation with human judgment is substantially explained by answer length, with a length-independent contribution not established on the current benchmark, as a genuine, disclosed limitation rather than omitting it. A paired, question-aware bootstrap contrast among the evaluator's three signals and their deployed composite (§X.B, Table VI) finds no statistically distinguished pairwise difference at this sample size, including between the composite and the CrossEncoder alone — a null result that reflects the benchmark's limited power as much as any genuine equivalence, and a question a properly powered, preregistered confirmatory study (§X.B) would need new data to settle. Future work includes: closing the gap between the follow-up mechanism's original five-condition design intent and its current implementation, or documenting the simpler mechanism as the intended design; a benchmark designed to separate reasoning quality from answer length (a protocol amendment for which is already drafted, pending institutional ethics clearance); a registered simulation evaluation of the specific deployed checkpoint; incorporating the project's companion PPO research once it is publicly available; and, contingent on ethics clearance and further governance decisions, a human feasibility/usability study. This paper makes no claim about candidate learning outcomes, which require direct evidence that does not yet exist.

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